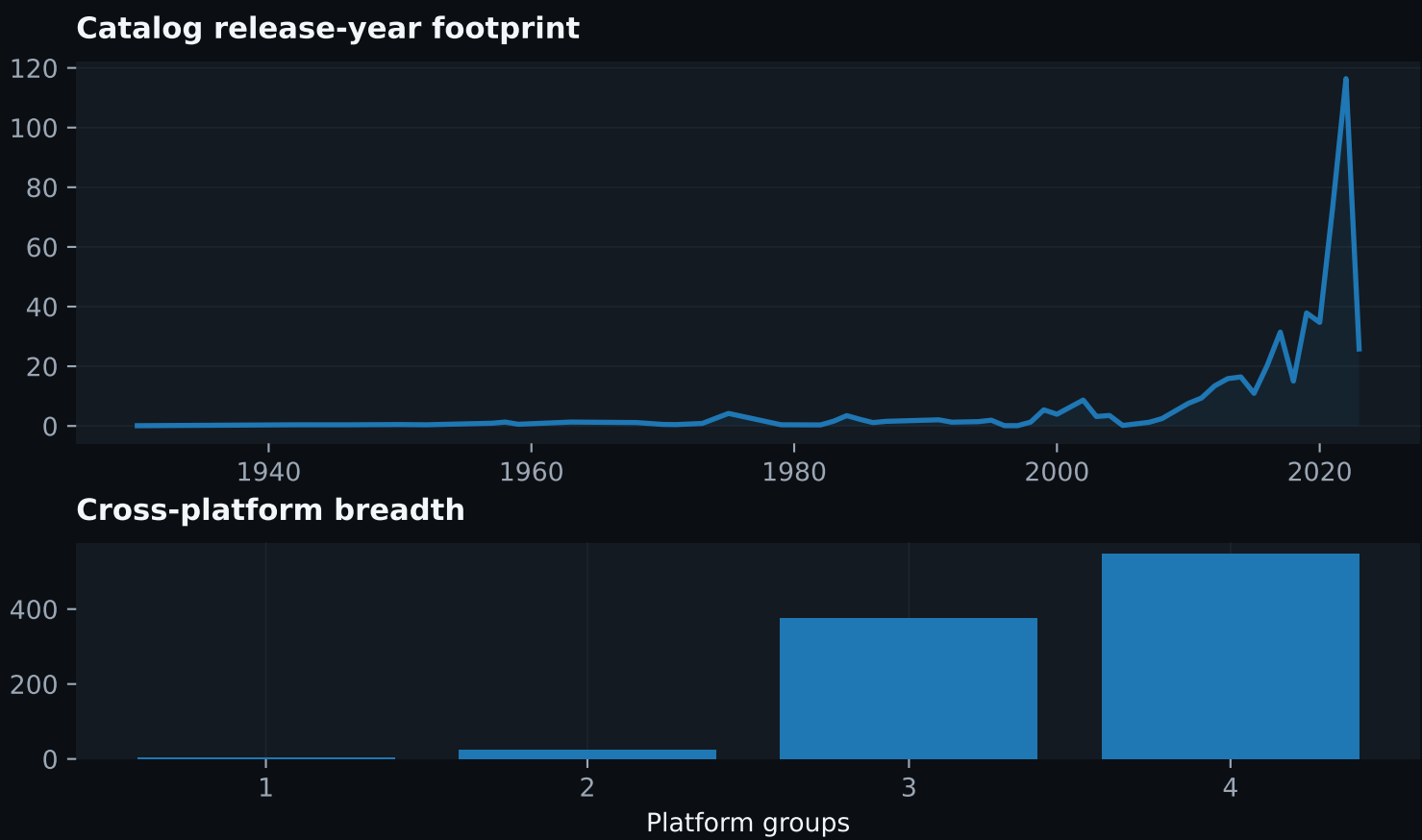
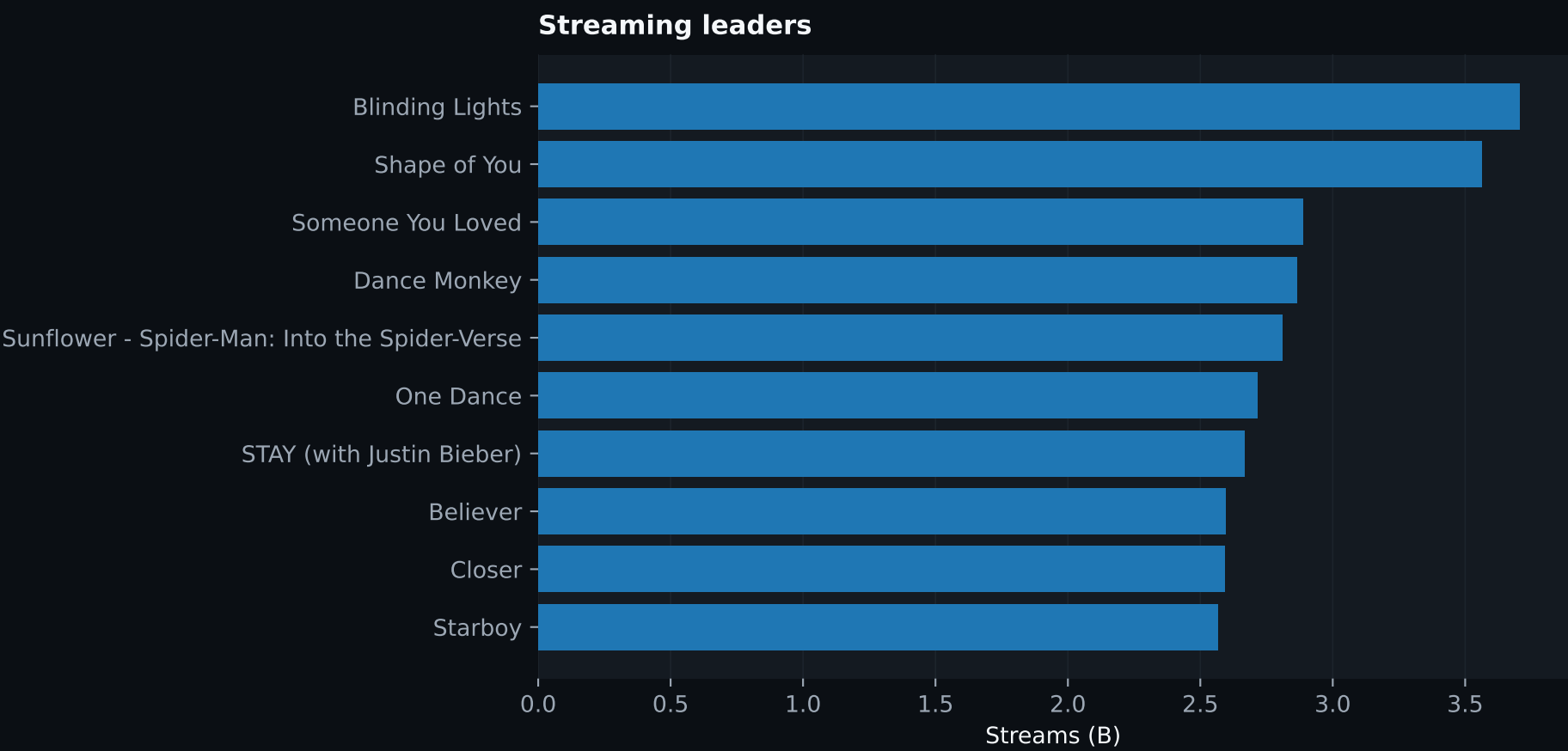


HOME	MUSIC	PLATFORMS	PLAYLISTS	CHARTS	OUTLIERS	ARTISTS	RESET FILTERS
TOTAL STREAMS 489.46B observed		SONG RECORDS 953 source rows		UNIQUE TRACKS 943 track names		ARTIST ENTRIES 645 as supplied	MEDIAN STREAMS 290.5M per record



MANAGEMENT INSIGHTS

- Playlist exposure is the strongest observed numerical relationship with streams ($r = 0.783$).
- Chart presence is much weaker ($r = 0.107$), so the two visibility concepts should not be treated as interchangeable.
- The report progresses from music characteristics to exposure, exceptions and artist concentration.

Music DNA | Audio Profile

Task 1A - Describe musical characteristics before testing popularity relationships.

TASK 1

02

HOME

MUSIC

PLATFORMS

PLAYLISTS

CHARTS

OUTLIERS

ARTISTS

RESET FILTERS

AVG BPM

122.54

DANCEABILITY

67.0%

ENERGY

64.3%

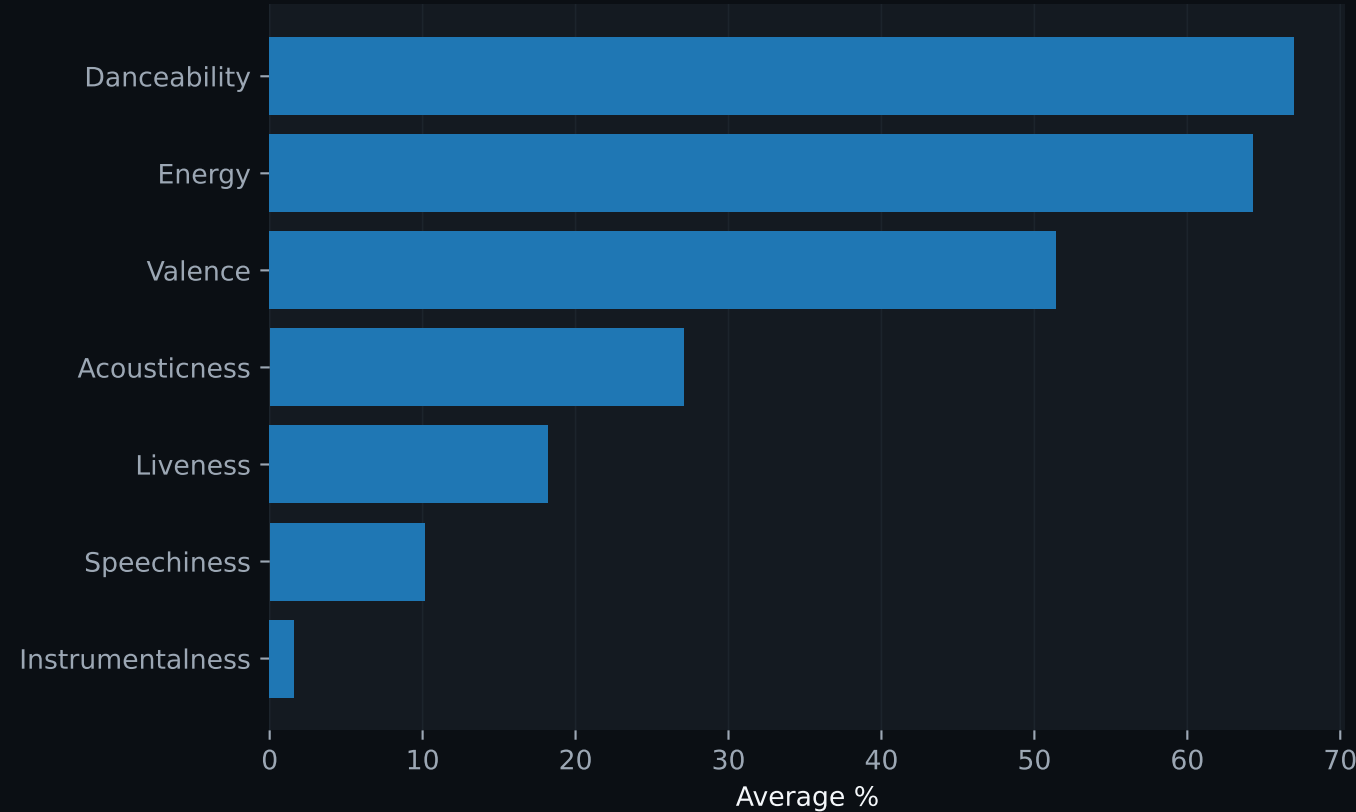
VALENCE

51.4%

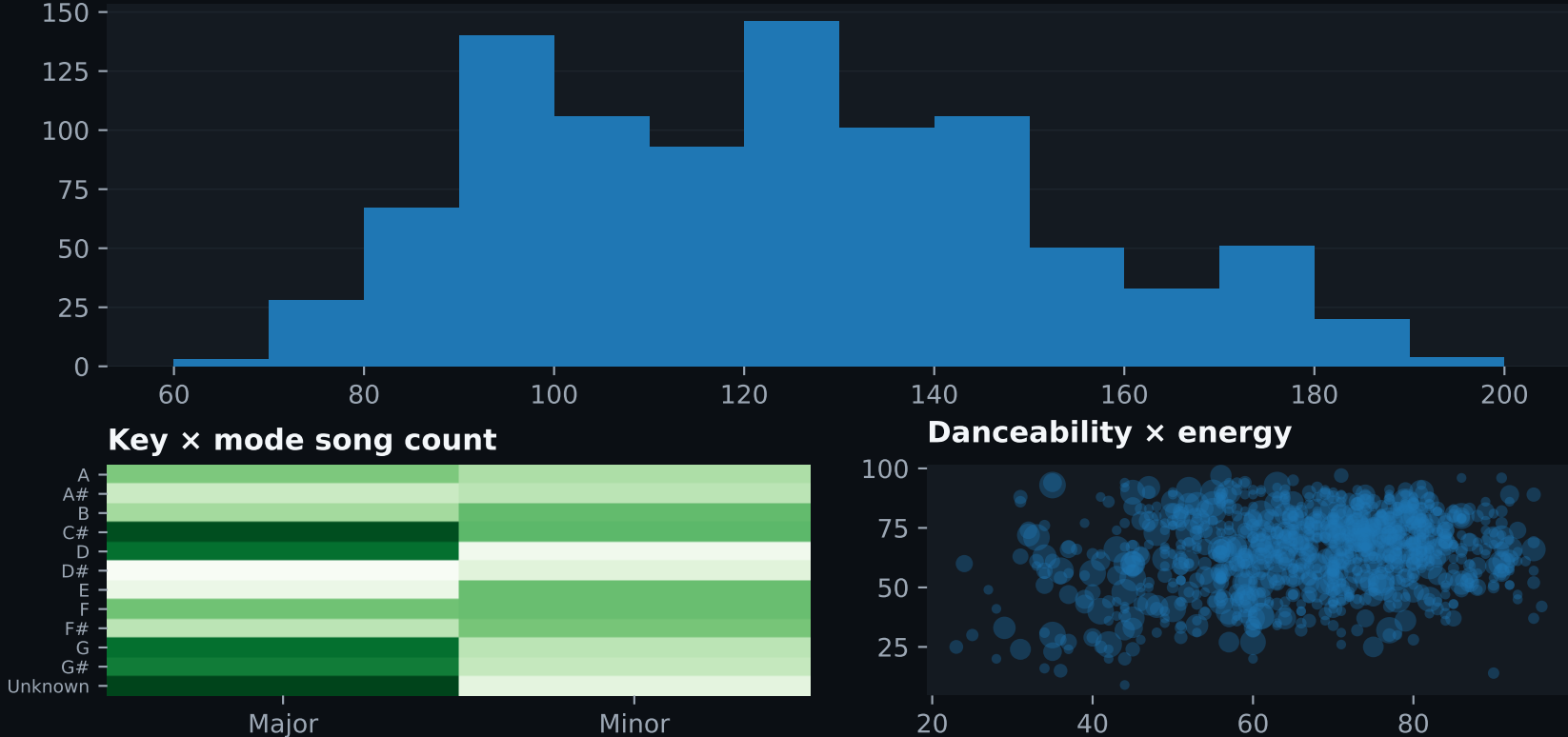
ACOUSTICNESS

27.1%

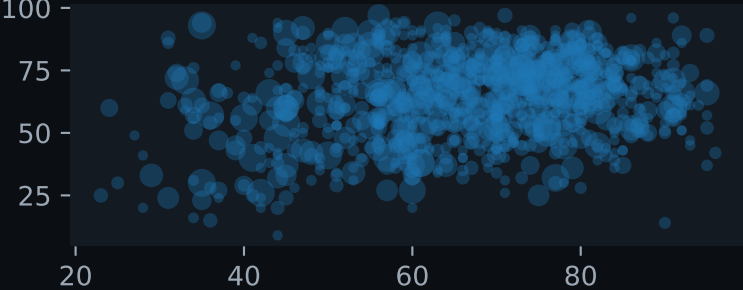
Average audio-feature profile



BPM distribution



Danceability x energy



MANAGEMENT INSIGHTS

• The average track is relatively danceable (67%) and energetic (64%), while average valence is more balanced at 51%.

• Instrumentalness is very low on average (1.6%), indicating that most records are not strongly instrumental by this measure.

• BPM is best interpreted as a distribution rather than a single target value.

Music DNA | Popularity Relationships

Task 1B - Test whether individual audio characteristics meaningfully explain streaming performance.

TASK 1

03

HOME

MUSIC

PLATFORMS

PLAYLISTS

CHARTS

OUTLIERS

ARTISTS

RESET FILTERS

STRONGEST |r|

0.112

DANCEABILITY r

-0.105

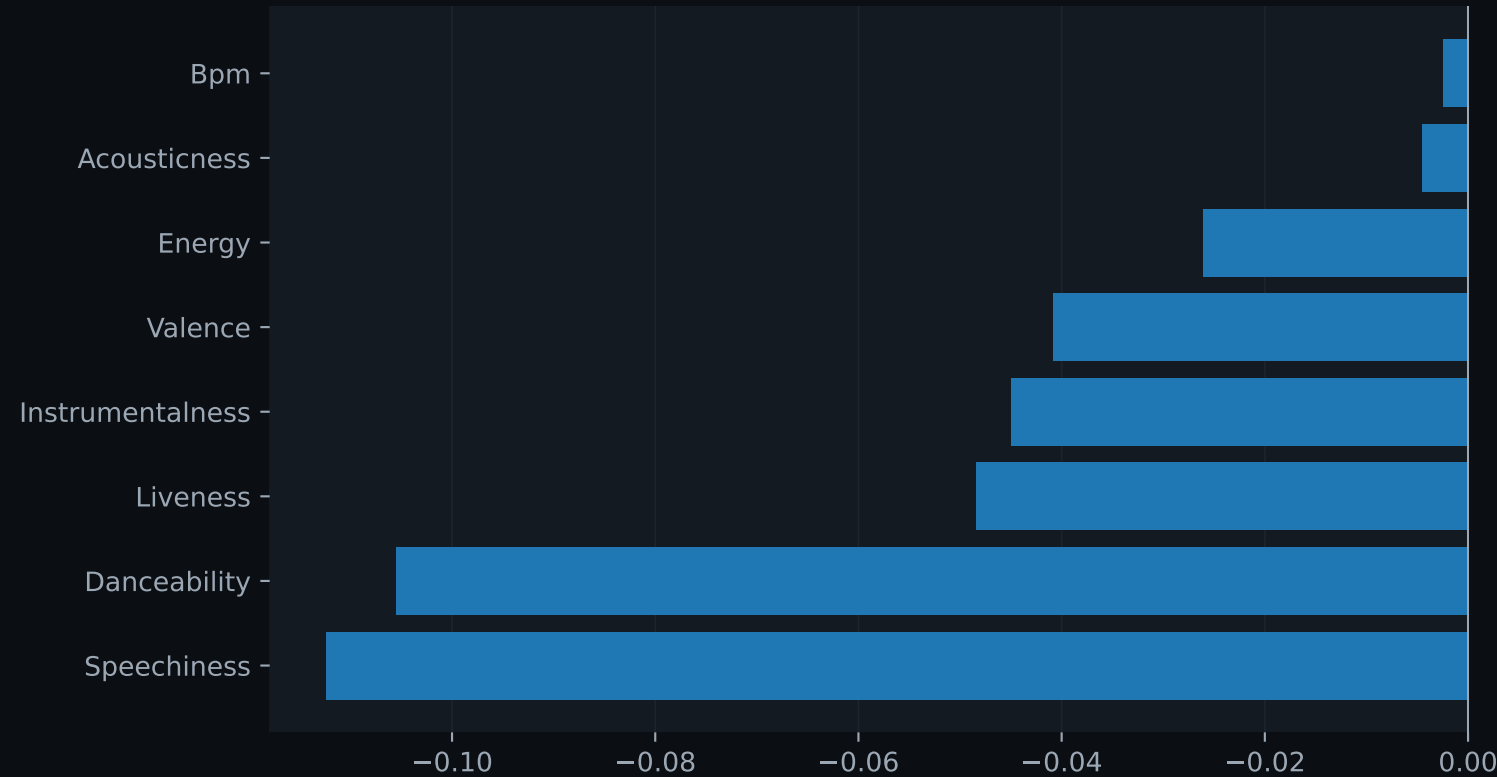
SPEECHINESS r

-0.112

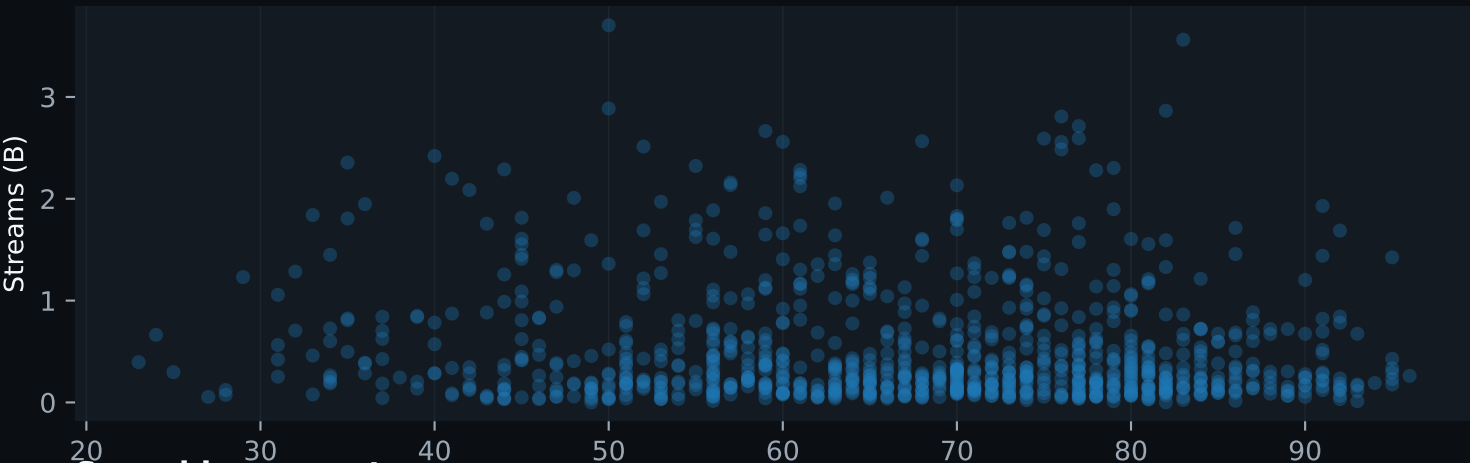
BPM r

-0.002

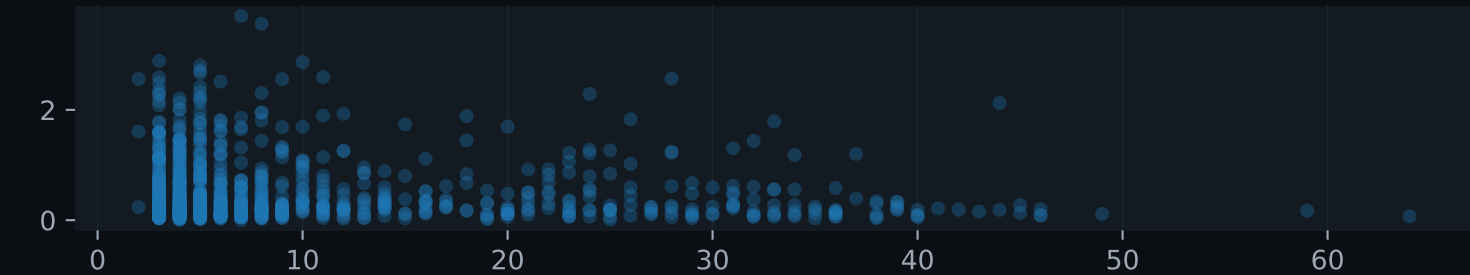
Correlation with streams



Danceability vs streams



Speechiness vs streams



MANAGEMENT INSIGHTS

• All individual audio-feature correlations with streams are weak in magnitude; the largest absolute value is only about 0.112.

• The data does not support a claim that one musical attribute drives streaming success.

• Use audio features for profile and segmentation, not causal popularity claims.

HOME

MUSIC

PLATFORMS

PLAYLISTS

CHARTS

OUTLIERS

ARTISTS

RESET FILTERS

4 PLATFORM GROUPS

549

3 GROUPS

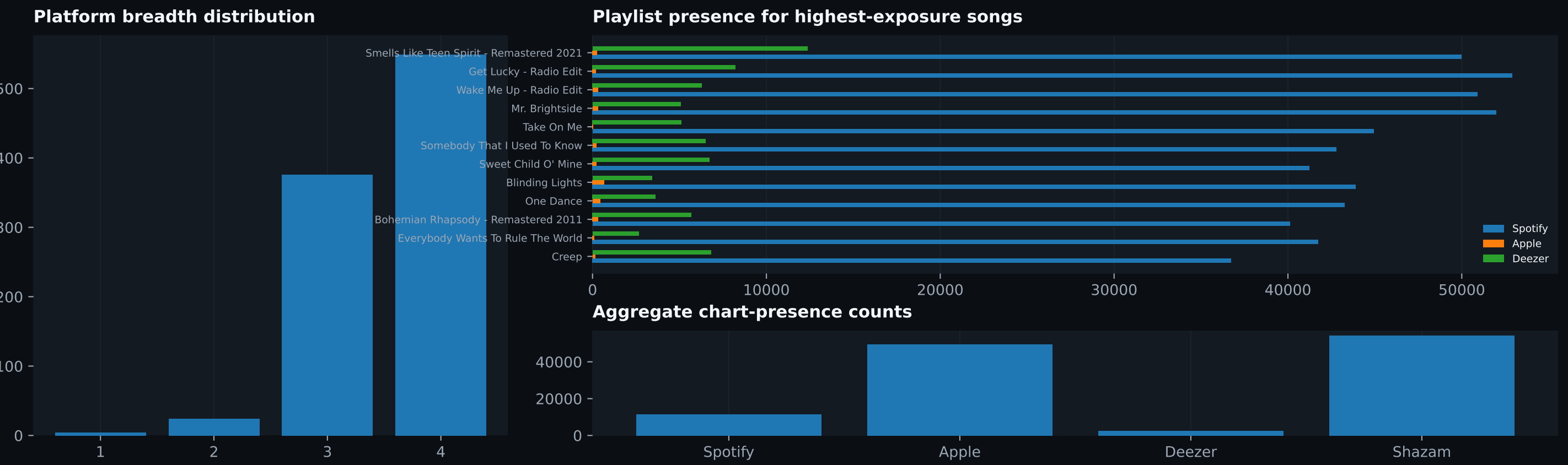
376

2 GROUPS

24

1 GROUP

4



Playlist Exposure | Relationship to Streaming

Task 3A - Quantify whether greater playlist exposure is associated with greater streaming totals.

HOME

MUSIC

PLATFORMS

PLAYLISTS

CHARTS

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ARTISTS

RESET FILTERS

CORRELATION

0.783

strong + association

MEDIAN EXPOSURE

2,308

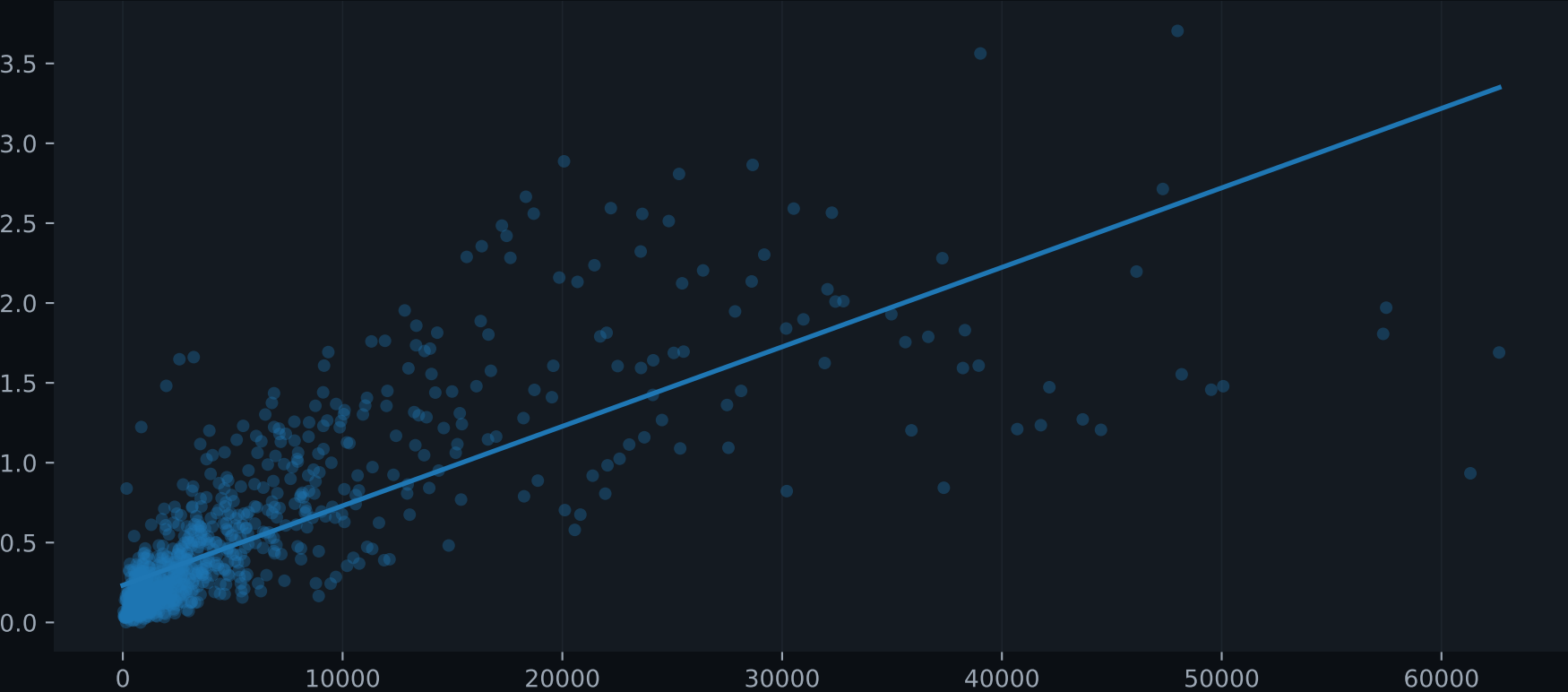
Q3 EXPOSURE

5,992

MEDIAN STREAMS

290.5M

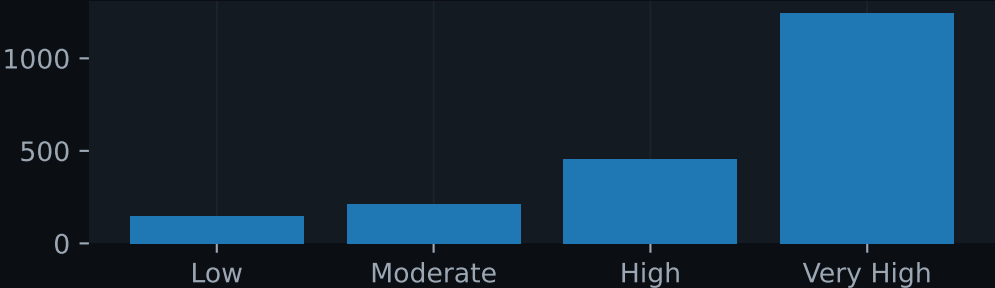
Playlist exposure vs streams



Exposure distribution



Average streams by exposure band



MANAGEMENT INSIGHTS

- Playlist exposure has a strong positive association with streams ($r = 0.783$).
- The exposure distribution is right-skewed, so median and percentile thresholds are more useful than the mean alone.
- The relationship is observational and should not be described as causal.

Playlist Exposure | Efficiency & Exceptions

Task 3B - Identify unusual streaming totals relative to observed playlist exposure.

HOME

MUSIC

PLATFORMS

PLAYLISTS

CHARTS

OUTLIERS

ARTISTS

RESET FILTERS

STREAM Q3

673.9M

EXPOSURE Q3

5,992

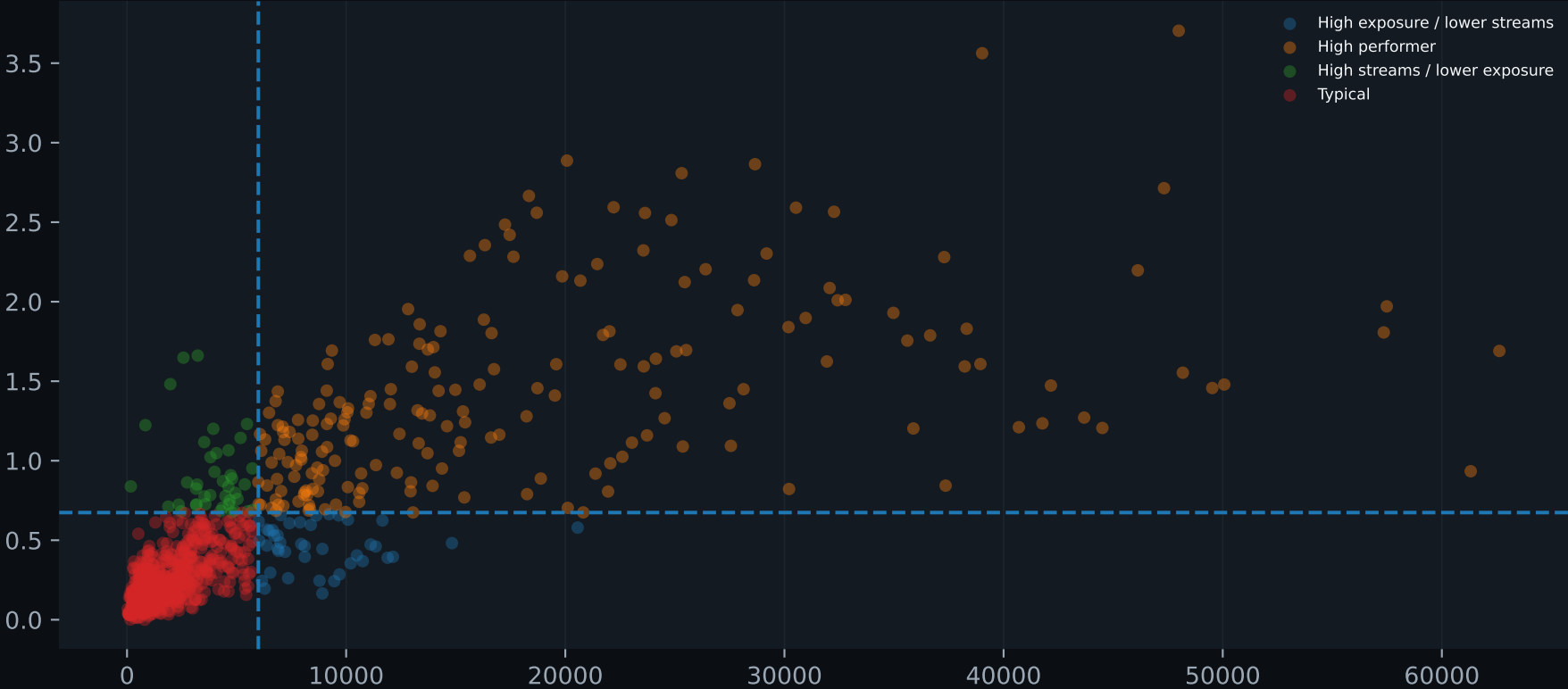
HIGH STREAM / LOWER EXP.

41

HIGH EXP. / LOWER STREAM

42

Exposure-performance quadrant



INVESTIGATION QUEUE

High streams / lower exposure

When I Was Your Man 1.66B
Die For You 1.65B
Locked Out Of Heaven 1.48B
traitor 1.23B
Enemy - from the series [...] 1.22B
Hope 1.20B

High exposure / lower streams

California Love - [...] 20,564
Do They Know It's [...] 14,827
White Christmas 12,136
The Christmas Song [...] 11,891
Sparks 11,655
Happy Xmas (War Is Over) 11,349

MANAGEMENT INSIGHTS

- Quadrant analysis converts correlation into an investigation workflow.
- Possible reasons for exceptions are outside the dataset and must not be presented as observed facts.
- Efficiency ratios are analytical indicators, not financial ROI.

HOME

MUSIC

PLATFORMS

PLAYLISTS

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ARTISTS

RESET FILTERS

PLAYLIST r

0.783

stronger

CHART r

0.107

weaker

TOTAL CHART PRESENCE

117,631

AVG CHART PRESENCE

123.4

Combined chart presence vs streams

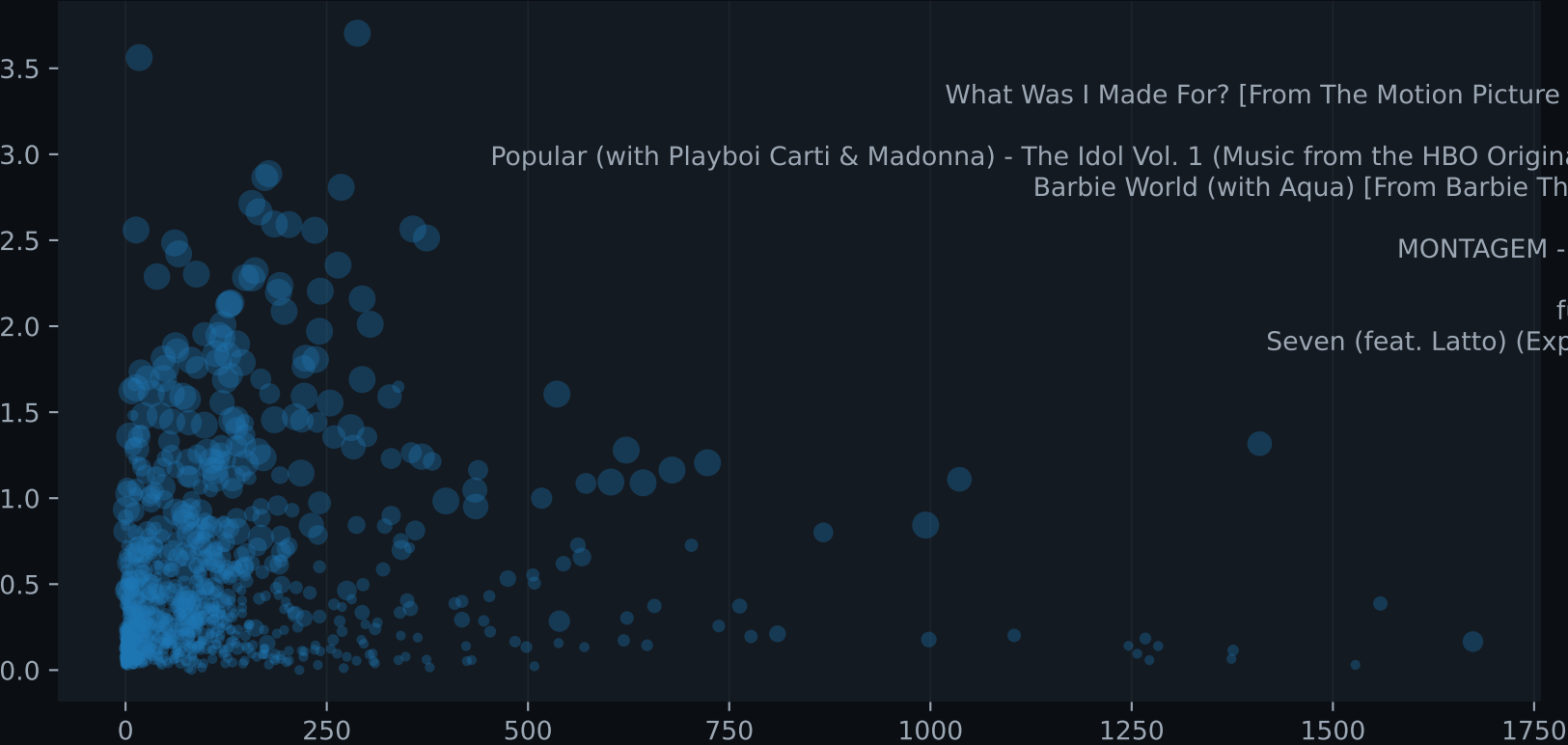
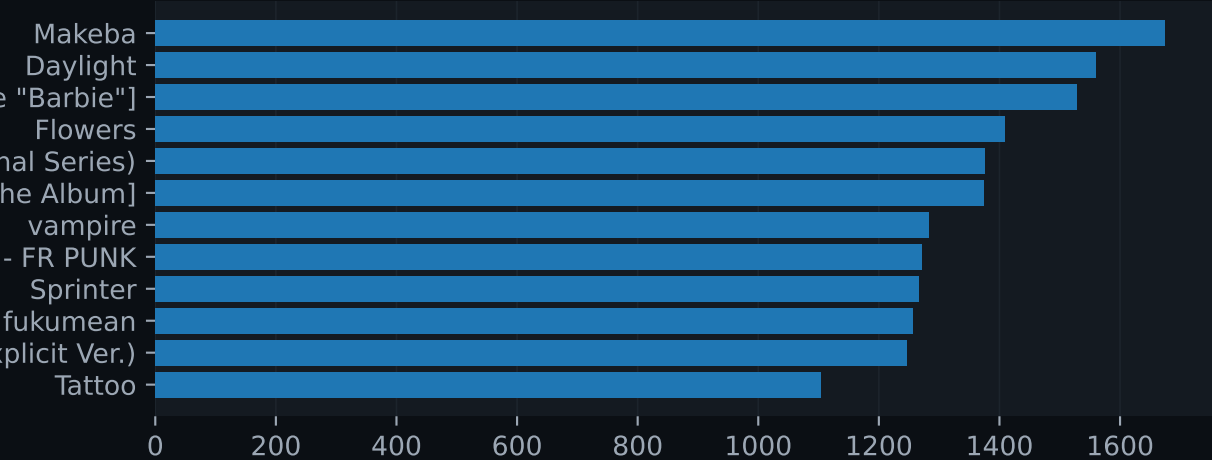
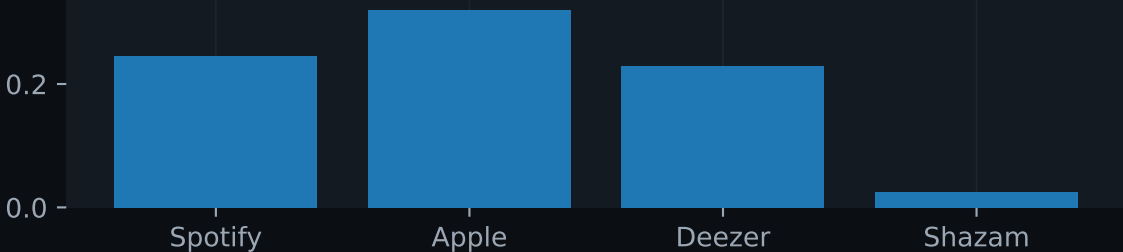


Chart-presence leaders



Platform chart correlation with streams



MANAGEMENT INSIGHTS

- Combined chart presence has a weak association with streams ($r = 0.107$), far below playlist exposure.
- Chart variables represent a different visibility signal and should not substitute for playlist exposure.
- Platform scale/timing differences limit direct normalization.

Outlier Intelligence | Detecting Songs That Break the Pattern

Task 5A - Separate typical songs, high performers and unusual exposure/stream combinations.

HOME MUSIC PLATFORMS PLAYLISTS CHARTS OUTLIERS ARTISTS RESET FILTERS

HIGH PERFORMER

197

HIGH STREAMS / LOWER EXPOSURE

41

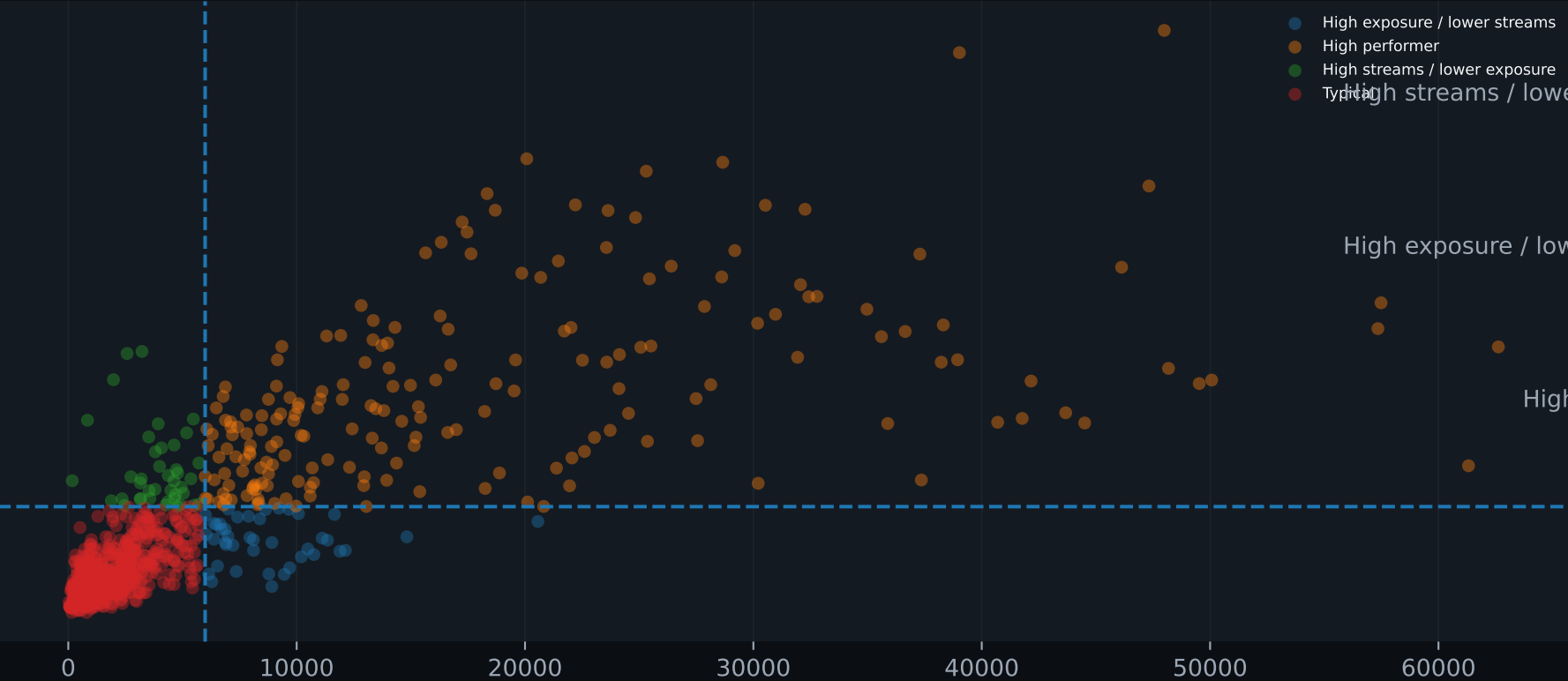
HIGH EXPOSURE / LOWER STREAMS

42

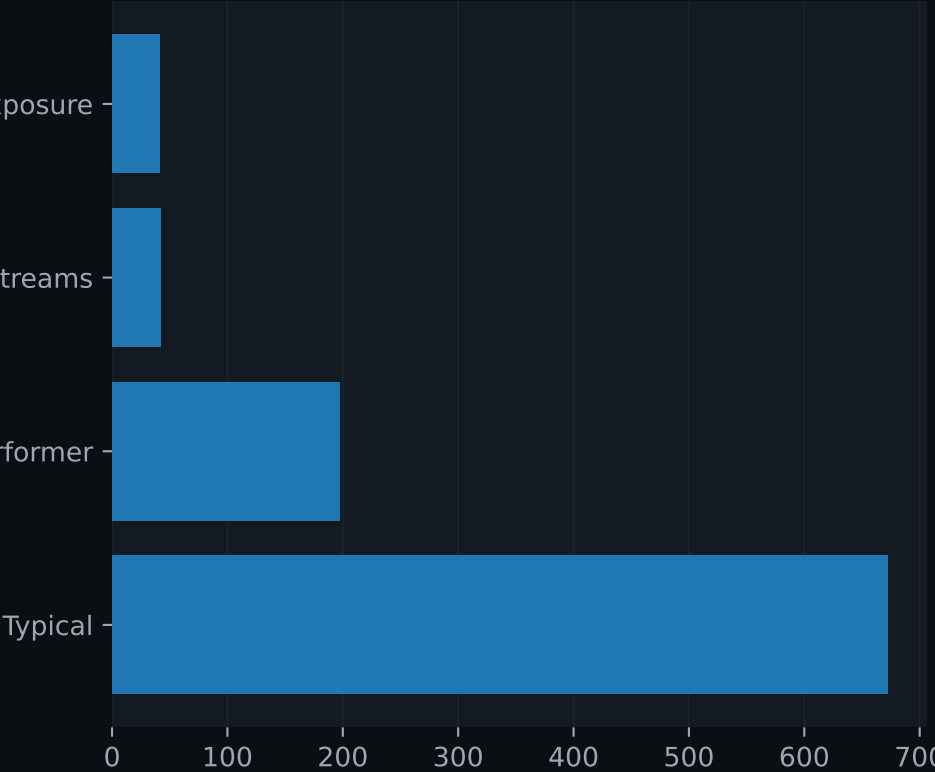
TYPICAL

672

Outlier classification map



Classification distribution



MANAGEMENT INSIGHTS

- Thresholds are empirical: Q3 streams = 673.87M and Q3 exposure = 5,992.
- Outlier classes are investigation categories, not quality judgments.
- Use drill-through to diagnose any selected song on the next page.

STREAMS

3.70B

PLAYLIST EXPOSURE

47,992

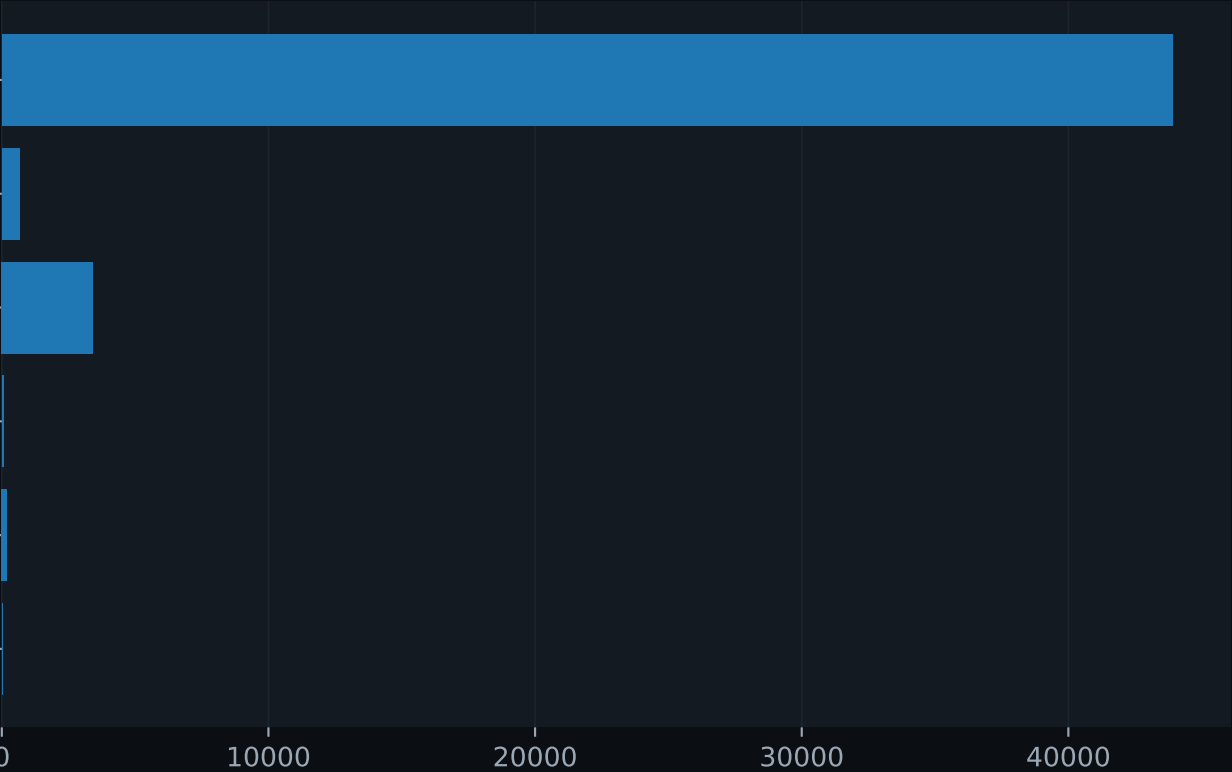
CHART PRESENCE

288

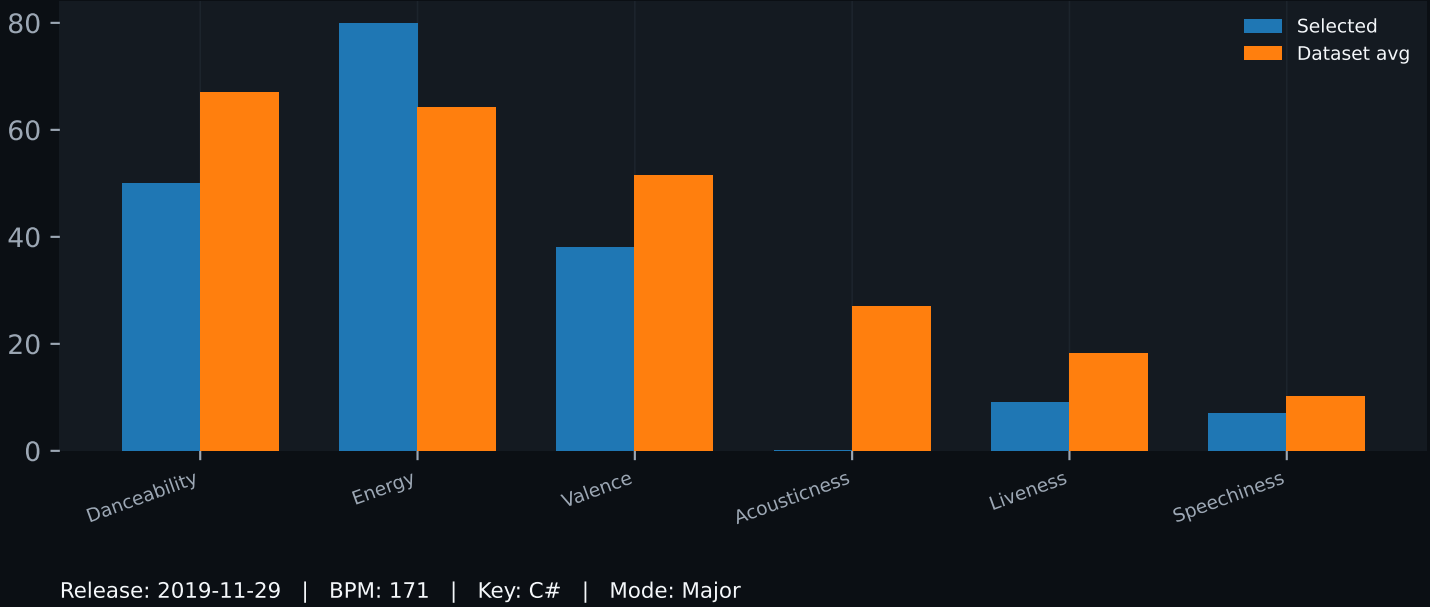
PLATFORM GROUPS

3

Selected song platform footprint



Audio profile vs dataset benchmark



MANAGEMENT INSIGHTS

- In Power BI, Track Name should be the drill-through field so this diagnostic works for any selected outlier.
- Benchmarking against dataset-average audio features separates musical profile from exposure and performance.
- No causal explanation is asserted because marketing, audience and external-media variables are absent.

Artist Concentration | Pareto Analysis

Task 6A - Quantify streaming concentration and test the 80/20 hypothesis.

TASK 6

10

HOME

MUSIC

PLATFORMS

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CHARTS

OUTLIERS

ARTISTS

RESET FILTERS

ARTIST ENTRIES

645

ARTISTS TO 80%

217

SHARE REQUIRED

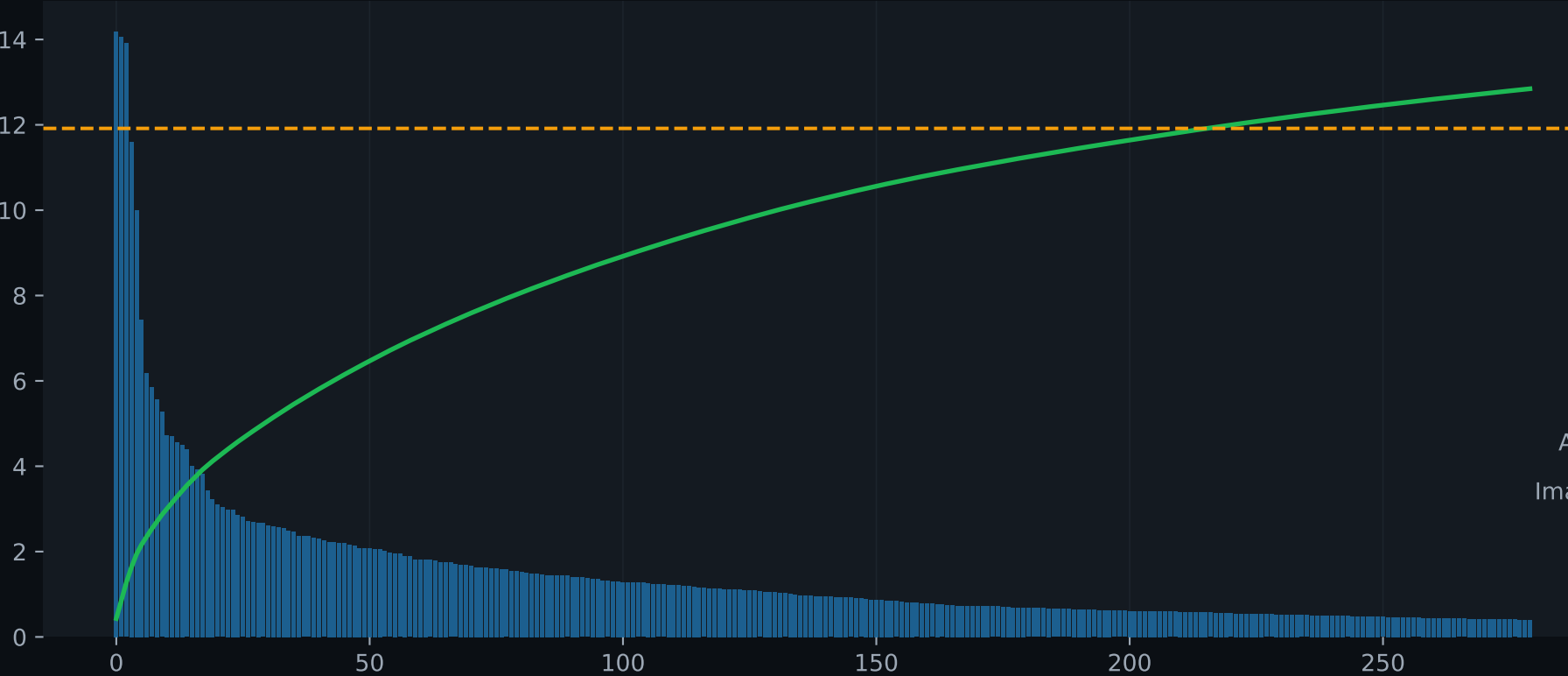
33.6%

not 20%

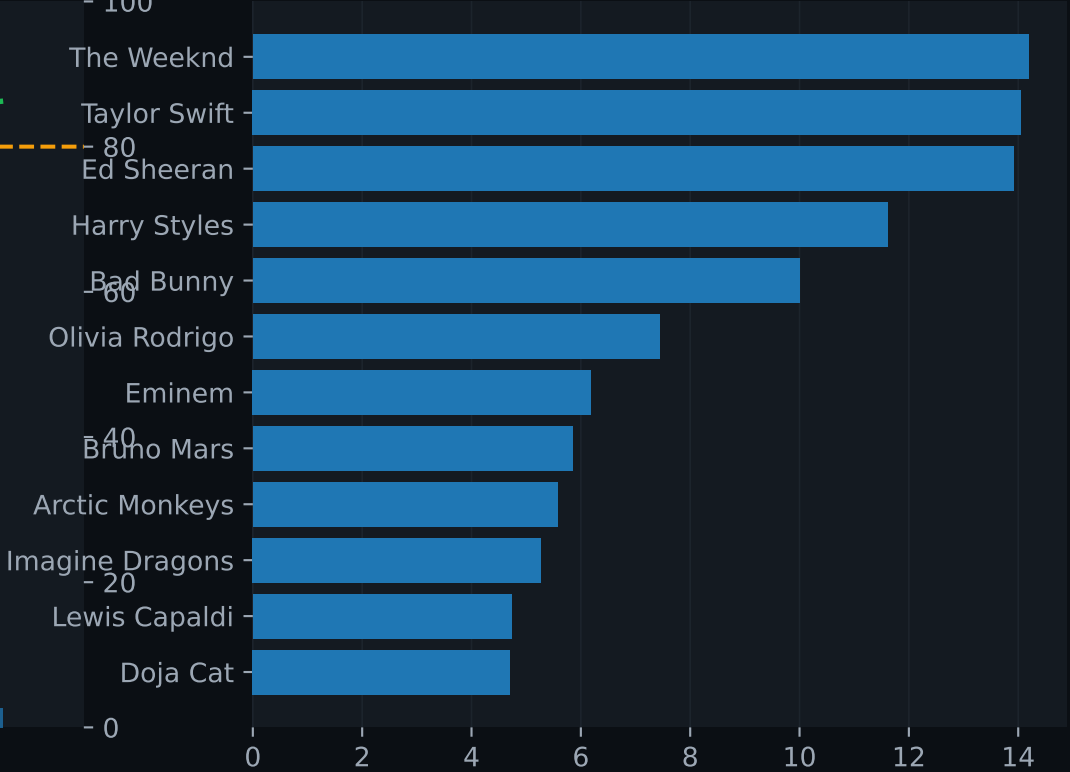
TOTAL STREAMS

489.46B

Pareto: ranked artist streams + cumulative share



Top artist entries



MANAGEMENT INSIGHTS

- 217 of 645 artist entries (33.6%) are needed to reach 80% of streams; this is not a strict 80/20 pattern.
- Streaming remains concentrated even though the classic Pareto ratio is not observed.
- Collaborative artist strings remain intact to preserve the source attribution.

Artist Portfolio | Scale, Breadth & Per-Track Strength

Task 6B - Distinguish catalog breadth from exceptional average track-level performance.

TASK 6

11

HOME

MUSIC

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OUTLIERS

ARTISTS

RESET FILTERS

THE WEEKND

14.19B

22 tracks

TAYLOR SWIFT

14.05B

34 tracks

ED SHEERAN

13.91B

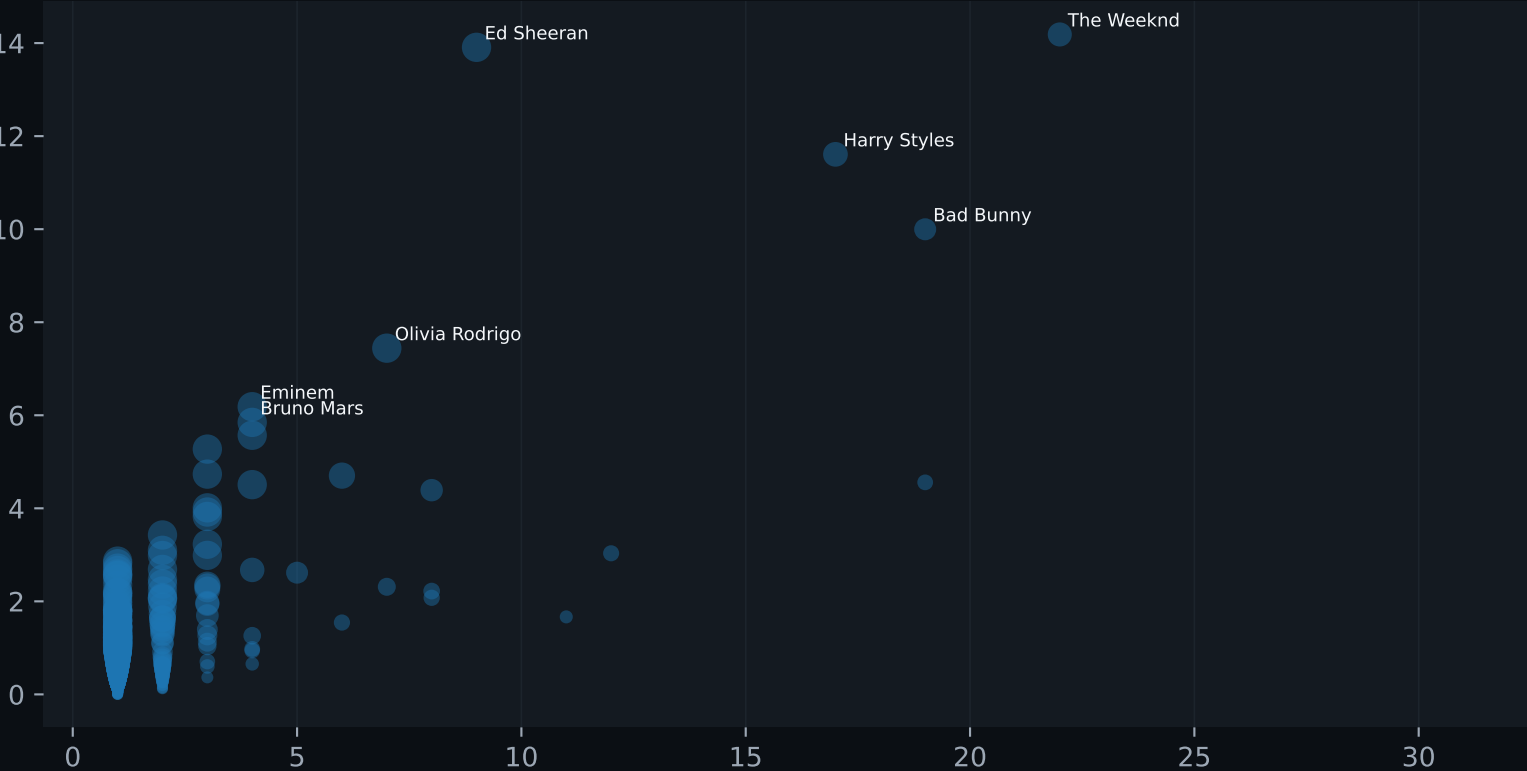
9 tracks

HARRY STYLES

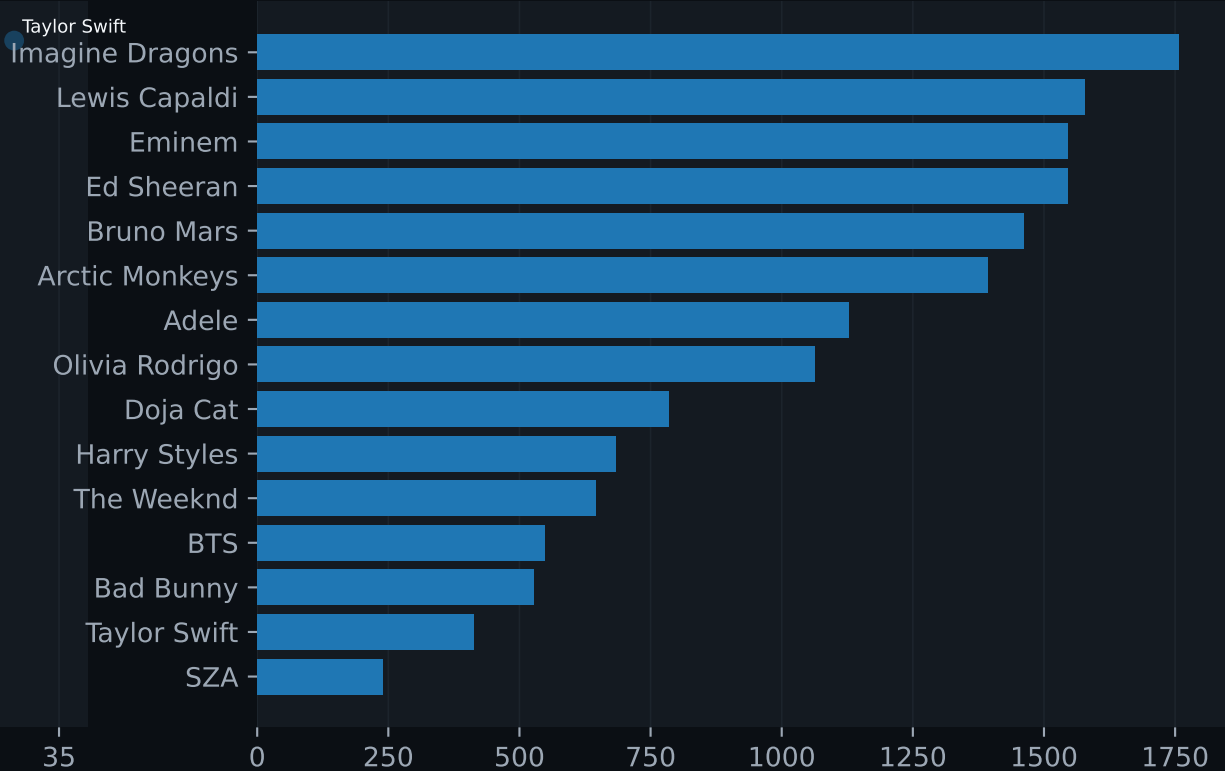
11.61B

17 tracks

Catalog breadth vs total streaming scale



Average streams per track | top total-stream artists



MANAGEMENT INSIGHTS

- Total artist streams should be read alongside track count; scale can come from breadth, hit strength, or both.
- Ed Sheeran reaches very high total streams with fewer represented tracks than Taylor Swift or The Weeknd.
- No artist-level revenue or audience metrics are inferred.

Model, Controls & Implementation Blueprint

Utility page - Data quality, model choices, slicer strategy, navigation and analytical guardrails.

HOME	MUSIC	PLATFORMS	PLAYLISTS	CHARTS	OUTLIERS	ARTISTS	RESET FILTERS
POWER QUERY	Convert streams to whole number; blank the one conversion error. Remove commas and convert Deezer playlists/Shazam charts to numeric. Create Release Date. Retain missing Key/Shazam values transparently.						
MODEL	Keep Spotify as the primary analytical table. Add a dedicated _Measures table. Add a Release Date dimension only if standardized date slicing is desired. Do not split collaborative artist strings unless attribution methodology changes.						
GLOBAL SLICERS	Artist and Release Year. Music pages add Key/Mode; platform pages add Platform Count; outlier pages add Classification.						
INTERACTIONS	Use cross-filtering selectively. Add Song and Artist report-page tooltips. Task 5 detail uses Track Name drill-through.						
NAVIGATION	Use Page Navigator for report sections. Create Reset_All bookmark with default slicer state. Add Back button on drill-through.						
GUARDRAILS	Release year is not streaming history. Platform variables use different scales. Correlation is not causation. No geography, revenue, demographics, genre or profitability exists in the source.						

MANAGEMENT INSIGHTS

- The model stays intentionally simple because the coursework has one analytical source table.
- Advanced features are used where they improve UX: drill-through, synchronized slicers, tooltips, bookmarks and navigation.
- Every analytical claim is constrained to fields available in the supplied dataset.